



Database Management Systems

Set Operations, Equivalent Queries

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Database Management Systems

Unit 1: Introduction to Database Management and SQL

Slides adapted from Author Slides of “Database System Concepts”, Silberschatz, H Korth and S Sudarshan, McGrawHill, 7th Edition, 2019. And Author Slides of Fundamentals of Database Systems”, Ramez Elamsri, Shamkant B Navathe, Pearson, 7th Edition, 2017.

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What are set operations ?

- Union Operation $\cup$
- Set-Intersection Operation $\cap$
- Set Difference Operation $-$

$\cup$

$\cap$

$-$



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Set Operations : UNION Operation



- Set Operations are Binary.
- These operations can be applied on the relations which are union compatible.
- Two relations are said to be union compatible if
 1. Both the relations have the *same arity* (number of attributes) and
 2. The corresponding attribute domains must be **compatible** (example: 2nd column of r should have the same type of values as does the 2nd column of s).

Union operation:

- The result of $R \cup S$, is a relation that includes all tuples that are either in R or in S or in both R and S . (R and S should be union compatible)
- Duplicate tuples are eliminated.
- The union operation allows us to combine two relations. **Notation:** $r \cup s$

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Sample University Database



Instructor Table:

ID	name	dept_name	salary
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000
33456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000
58583	Califieri	History	62000
76543	Singh	Finance	80000
76766	Crick	Biology	72000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000

Course Table:

course_id	title	dept_name	credits
BIO-101	Intro. to Biology	Biology	4
BIO-301	Genetics	Biology	4
BIO-399	Computational Biology	Biology	3
CS-101	Intro. to Computer Science	Comp. Sci.	4
CS-190	Game Design	Comp. Sci.	4
CS-315	Robotics	Comp. Sci.	3
CS-319	Image Processing	Comp. Sci.	3
CS-347	Database System Concepts	Comp. Sci.	3
EE-181	Intro. to Digital Systems	Elec. Eng.	3
FIN-201	Investment Banking	Finance	3
HIS-351	World History	History	3
MU-199	Music Video Production	Music	3
PHY-101	Physical Principles	Physics	4

Prerequisites Table:

course_id	prereq_id
BIO-301	BIO-101
BIO-399	BIO-101
CS-190	CS-101
CS-315	CS-101
CS-319	CS-101
CS-347	CS-101
EE-181	PHY-101

Teaches Table:

Department Table:

dept_name	building	budget
Biology	Watson	90000
Comp. Sci.	Taylor	100000
Elec. Eng.	Taylor	85000
Finance	Painter	120000
History	Painter	50000
Music	Packard	80000
Physics	Watson	70000

Figure 2.5 The department relation.

Section Table:

course_id	sec_id	semester	year	building	room_number	time_slot_id
BIO-101	1	Summer	2017	Painter	514	B
BIO-301	1	Summer	2018	Painter	514	A
CS-101	1	Fall	2017	Packard	101	H
CS-101	1	Spring	2018	Packard	101	F
CS-190	1	Spring	2017	Taylor	3128	E
CS-190	2	Spring	2017	Taylor	3128	A
CS-315	1	Spring	2018	Watson	120	D
CS-319	1	Spring	2018	Watson	100	B
CS-319	2	Spring	2018	Taylor	3128	C
CS-347	1	Fall	2017	Taylor	3128	A
EE-181	1	Spring	2017	Taylor	3128	C
FIN-201	1	Spring	2018	Packard	101	B
HIS-351	1	Spring	2018	Painter	514	C
MU-199	1	Spring	2018	Packard	101	D
PHY-101	1	Fall	2017	Watson	100	A

ID	course_id	sec_id	semester	year
10101	CS-101	1	Fall	2017
10101	CS-315	1	Spring	2018
10101	CS-347	1	Fall	2017
12121	FIN-201	1	Spring	2018
15151	MU-199	1	Spring	2018
22222	PHY-101	1	Fall	2017
32343	HIS-351	1	Spring	2018
45565	CS-101	1	Spring	2018
45565	CS-319	1	Spring	2018
76766	BIO-101	1	Summer	2017
76766	BIO-301	1	Summer	2018
83821	CS-190	1	Spring	2017
83821	CS-190	2	Spring	2017
83821	CS-319	2	Spring	2018
98345	EE-181	1	Spring	2017

Figure 2.7 The teaches relation.

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Set Operations : UNION Operation



Schema:

<i>course_id</i>	<i>sec_id</i>	<i>semester</i>	<i>year</i>	<i>building</i>	<i>room_number</i>	<i>time_slot_id</i>
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Example: Find all courses (*course_id*) taught in the Fall 2017 semester, or in the Spring 2018 semester, or in both

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Set Operations : UNION Operation



Query:

$$\Pi_{course_id} (\sigma_{semester="Fall" \wedge year=2017}(section)) \cup \Pi_{course_id} (\sigma_{semester="Spring" \wedge year=2018}(section))$$

Table:

course_id	sec_id	semester	year	building	room_number	time_slot_id
BIO-101	1	Summer	2017	Painter	514	B
BIO-301	1	Summer	2018	Painter	514	A
CS-101	1	Fall	2017	Packard	101	H
CS-101	1	Spring	2018	Packard	101	F
CS-190	1	Spring	2017	Taylor	3128	E
CS-190	2	Spring	2017	Taylor	3128	A
CS-315	1	Spring	2018	Watson	120	D
CS-319	1	Spring	2018	Watson	100	B
CS-319	2	Spring	2018	Taylor	3128	C
CS-347	1	Fall	2017	Taylor	3128	A
EE-181	1	Spring	2017	Taylor	3128	C
FIN-201	1	Spring	2018	Packard	101	B
HIS-351	1	Spring	2018	Painter	514	C
MU-199	1	Spring	2018	Packard	101	D
PHY-101	1	Fall	2017	Watson	100	A

Output:

course_id
CS-101
CS-315
CS-319
CS-347
FIN-201
HIS-351
MU-199
PHY-101

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Set Operations : UNION Operation

What will be the first Column Name after Union of Paternity and Maternity Relation?

Is It Meaningful?

Table:

Paternity

Father	Child
Adam	Cain
Adam	Abel
Abraham	Isaac
Abraham	Ishmael

Maternity

Mother	Child
Eve	Cain
Eve	Seth
Sarah	Isaac
Hagar	Ishmael

Paternity $\cup$ Maternity ???

The problem: **Father** and **Mother** are different names, but both represent a parent.

Output:

Renaming and Union

Paternity

Father	Child
Adam	Cain
Adam	Abel
Abraham	Isaac
Abraham	Ishmael

Maternity

Mother	Child
Eve	Cain
Eve	Seth
Sarah	Isaac
Hagar	Ishmael

$\rho_{\text{Father} \rightarrow \text{Parent}}(\text{Paternity}) \cup \rho_{\text{Mother} \rightarrow \text{Parent}}(\text{Maternity})$

Parent	Child
Adam	Cain
Adam	Abel
Abraham	Isaac
Abraham	Ishmael
Eve	Cain
Eve	Seth
Sarah	Isaac
Hagar	Ishmael

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Set Operations : UNION Operation



- To retrieve the social security numbers of all employees who either work in department 5 (RESULT1 below) or directly supervise an employee who works in department 5 (RESULT2 below).
- We can use the UNION operation as follows:

```
DEP5_EMPS  $\leftarrow$   $\sigma_{DNO=5}$  (EMPLOYEE)
RESULT1  $\leftarrow$   $\pi_{SSN}$ (DEP5_EMPS)
RESULT2(SSN)  $\leftarrow$   $\pi_{SUPERSSN}$ (DEP5_EMPS)
RESULT  $\leftarrow$  RESULT1  $\cup$  RESULT2
```

- The union operation produces the tuples that are in either RESULT1 or RESULT2 or both.

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Set Operations : UNION Operation

RESULT $\leftarrow$ RESULT1 $\cup$ RESULT2.

RESULT1

Ssn
123456789
333445555
666884444
453453453

RESULT2

Ssn
333445555
888665555

RESULT

Ssn
123456789
333445555
666884444
453453453
888665555

- The set-intersection operation allows us to find tuples that are in both the input relations.

Notation: $r \cap s$

- r, s have the *same arity* attributes of r and s are **compatible**

Example: Find the set of all courses taught in both the **Fall 2017** and the **Spring 2018** semesters.

Query: $\pi_{course_id} (\sigma_{semester="Fall" \wedge year=2017} (section)) \cap \pi_{course_id} (\sigma_{semester="Spring" \wedge year=2018} (section))$

course_id	sec_id	semester	year	building	room_number	time_slot_id
BIO-101	1	Summer	2017	Painter	514	B
BIO-301	1	Summer	2018	Painter	514	A
CS-101	1	Fall	2017	Packard	101	H
CS-101	1	Spring	2018	Packard	101	F
CS-190	1	Spring	2017	Taylor	3128	E
CS-190	2	Spring	2017	Taylor	3128	A
CS-315	1	Spring	2018	Watson	120	D
CS-319	1	Spring	2018	Watson	100	B
CS-319	2	Spring	2018	Taylor	3128	C
CS-347	1	Fall	2017	Taylor	3128	A
EE-181	1	Spring	2017	Taylor	3128	C
FIN-201	1	Spring	2018	Packard	101	B
HIS-351	1	Spring	2018	Painter	514	C
MU-199	1	Spring	2018	Packard	101	D
PHY-101	1	Fall	2017	Watson	100	A

Output:

course_id
CS-101

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Set Operations : Intersection Operation

Example : Graduates $\cap$ Managers

Graduates

Number	Surname	Age
7274	Robinson	37
7432	O'Malley	39
9824	Darkes	38

Managers

Number	Surname	Age
9297	O'Malley	58
7432	O'Malley	39
9824	Darkes	38

Output:

Graduates $\cap$ Managers

Number	Surname	Age
7432	O'Malley	39
9824	Darkes	38

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Set Operations : SET DIFFERENCE



The set-difference operation allows us to find tuples that are in one relation but are not in another.

Notation: $r - s$

Set differences must be applied on **compatible** relations ie r and s must have the same **arity** and attribute domains of r and s must be **compatible**.

Example: To find all courses taught in the Fall 2017 semester, but not in the Spring 2018 semester.

Query: $\Pi_{course_id} (\sigma_{semester="Fall" \wedge year=2017}(section)) - \Pi_{course_id} (\sigma_{semester="Spring" \wedge year=2018}(section))$

Output:

<i>course_id</i>
CS-347
PHY-101

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Set Operations : Intersection Operation

Example : Graduates - Managers

Graduates

Number	Surname	Age
7274	Robinson	37
7432	O'Malley	39
9824	Darkes	38

Managers

Number	Surname	Age
9297	O'Malley	56
7432	O'Malley	39
9824	Darkes	38

Output:

Graduates - Managers

Number	Surname	Age
7274	Robinson	37

Example : STUDENT U INSTRUCTOR

STUDENT

Fn	Ln
Susan	Yao
Ramesh	Shah
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert

INSTRUCTOR

Fname	Lname
John	Smith
Ricardo	Browne
Susan	Yao
Francis	Johnson
Ramesh	Shah

Output:

Fn	Ln
Susan	Yao
Ramesh	Shah
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert
John	Smith
Ricardo	Browne
Francis	Johnson

Example : $\text{STUDENT} \cap \text{INSTRUCTOR}$.

STUDENT

Fn	Ln
Susan	Yao
Ramesh	Shah
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert

INSTRUCTOR

Fname	Lname
John	Smith
Ricardo	Browne
Susan	Yao
Francis	Johnson
Ramesh	Shah

Output:

Fn	Ln
Susan	Yao
Ramesh	Shah

Example : STUDENT – INSTRUCTOR.

STUDENT

Fn	Ln
Susan	Yao
Ramesh	Shah
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert

INSTRUCTOR

Fname	Lname
John	Smith
Ricardo	Browne
Susan	Yao
Francis	Johnson
Ramesh	Shah

Output:

Fn	Ln
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert

Example : INSTRUCTOR – STUDENT.

STUDENT

Fn	Ln
Susan	Yao
Ramesh	Shah
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert

INSTRUCTOR

Fname	Lname
John	Smith
Ricardo	Browne
Susan	Yao
Francis	Johnson
Ramesh	Shah

Output:

Fname	Lname
John	Smith
Ricardo	Browne
Francis	Johnson

Notice that both union and intersection are **commutative** operations; that is:

- $R \cup S = S \cup R$ and $R \cap S = S \cap R$

Both union and intersection can be treated as **n-ary** operations applicable to any number of relations as both are **associative** operations; that is:

- $R \cup (S \cup T) = (R \cup S) \cup T$
- $(R \cap S) \cap T = R \cap (S \cap T)$

The minus operation is not commutative; that is, in general:

- $R - S \neq S - R$

- The Division operation is also a binary operation.
- $R(Z) \div S(X)$, where X is a subset of Z .
- $Y = Z - X$ that is let Y is the set of attributes of R that are not attributes of S in the resultant relation (T).
- The result of DIVISION is a relation $T(Y)$ that includes a tuple t if tuples tR appear in R with $tR[Y] = t$, and with $tR[X] = ts$ for every tuple ts in S .
- For a tuple t to appear in the result T of the DIVISION, the values in t must appear in R in combination with every tuple in S .

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Set Operations : Division Operation

Example:

(a) Dividing SSN_PNOS by SMITH_PNOS.

(a)

SSN_PNOS

Essn	Pno
123456789	1
123456789	2
666884444	3
453453453	1
453453453	2
333445555	2
333445555	3
333445555	10
333445555	20
999887777	30
999887777	10
987987987	10
987987987	30
987654321	30
987654321	20
888665555	20

SMITH_PNOS

Pno
1
2

SSNS

Ssn
123456789
453453453

(b) $T \leftarrow R \div S$

(b)

R

A	B
a1	b1
a2	b1
a3	b1
a4	b1
a1	b2
a3	b2
a2	b3
a3	b3
a4	b3
a1	b4
a2	b4
a3	b4

S

A
a1
a2
a3

T

B
b1
b4

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Equivalent Queries



There is more than one way to write a query in relational algebra.

Example:

Find information about courses taught by instructors in the Physics department with salary greater than 90,000.

Query 1: $\sigma_{dept_name="Physics"} \wedge salary > 90,000 (instructor)$

Query 2 : $\sigma_{dept_name="Physics"} (\sigma_{salary > 90,000} (instructor))$

The two queries are not identical; they are however equivalent -- they give the same result on the given database.

- In Relational Algebra, the Cartesian Product ($\times$) is a binary operation that combines every tuple of one relation with every tuple of another relation.
- Allows to combine information from two relations.
- Denoted as **$R \times S$**
- n - number of attributes in R and m - number of attributes in S
- Result is a relation Q with degree (arity) $n + m$ attributes and $Q(A_1, A_2, \dots, A_n, B_1, B_2, \dots, B_m)$ order of attributes.
- The resulting relation state has one tuple for each combination of tuples—one from R and one from S .
- Hence, if R has n_R tuples (denoted as $|R| = n_R$), and S has n_S tuples, then $R \times S$ will have $n_R * n_S$ tuples.
- The two operands do NOT have to be "union compatible"

Sample Employee Table

Fname	Minit	Lname	Ssn	Bdate	Address	Gender	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren,Houston,TX	M	30000.00	888665555	5
Franklin	T	Wong	333445555	1955-12-08	638 voss,Houston,TX	M	40000.00	888665555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice,Houston,TX	F	25000.00	333445555	5
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000.00	333445555	5
James	E	Borg	888665555	1937-11-10	450 Stone, Houston,TX	M	55000.00	NULL	1
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire,Tx	F	43000.00	333445555	4
Ahmed	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston,TX	M	25000.00	987654321	4
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring,Tx	F	25000.00	333445555	4

Sample Dependent Table

Essn	Dependent_name	Gender	Bdate	Relationship
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse
123456789	Michael	M	1988-01-04	Son
333445555	Alice	F	1986-04-05	Daughter
333445555	Joy	F	1958-05-03	Spouse
333445555	Theodore	M	1983-10-25	Son
987654321	Abner	M	1942-02-28	Spouse

Sample Cartesian Product : Employee X Dependent

Fname	Minit	Lname	Ssn	Bdate	Address	Gender	Salary	Super_ssn	Dno	Essn	Dependent_name	Gender	Bdate	Relati
John	B	Smith	123456789	1965-01-09	731 Fondren,Houston,TX	M	30000.00	888665555	5	987654321	Abner	M	1942-02-28	Spouse
John	B	Smith	123456789	1965-01-09	731 Fondren,Houston,TX	M	30000.00	888665555	5	333445555	Theodore	M	1983-10-25	Son
John	B	Smith	123456789	1965-01-09	731 Fondren,Houston,TX	M	30000.00	888665555	5	333445555	Joy	F	1958-05-03	Spouse
John	B	Smith	123456789	1965-01-09	731 Fondren,Houston,TX	M	30000.00	888665555	5	333445555	Alice	F	1986-04-05	Daught
John	B	Smith	123456789	1965-01-09	731 Fondren,Houston,TX	M	30000.00	888665555	5	123456789	Michael	M	1988-01-04	Son
John	B	Smith	123456789	1965-01-09	731 Fondren,Houston,TX	M	30000.00	888665555	5	123456789	Elizabeth	F	1967-05-05	Spouse
John	B	Smith	123456789	1965-01-09	731 Fondren,Houston,TX	M	30000.00	888665555	5	123456789	Alice	F	1988-12-30	Daught
Franklin	T	Wong	333445555	1955-12-08	638 voss,Houston,TX	M	40000.00	888665555	5	987654321	Abner	M	1942-02-28	Spouse
Franklin	T	Wong	333445555	1955-12-08	638 voss,Houston,TX	M	40000.00	888665555	5	333445555	Theodore	M	1983-10-25	Son
Franklin	T	Wong	333445555	1955-12-08	638 voss,Houston,TX	M	40000.00	888665555	5	333445555	Joy	F	1958-05-03	Spouse
Franklin	T	Wong	333445555	1955-12-08	638 voss,Houston,TX	M	40000.00	888665555	5	333445555	Alice	F	1986-04-05	Daught
Franklin	T	Wong	333445555	1955-12-08	638 voss,Houston,TX	M	40000.00	888665555	5	123456789	Michael	M	1988-01-04	Son
Franklin	T	Wong	333445555	1955-12-08	638 voss,Houston,TX	M	40000.00	888665555	5	123456789	Elizabeth	F	1967-05-05	Spouse
Franklin	T	Wong	333445555	1955-12-08	638 voss,Houston,TX	M	40000.00	888665555	5	123456789	Alice	F	1988-12-30	Daught
Joyce	A	English	453453453	1972-07-31	5631 Rice,Houston,TX	F	25000.00	333445555	5	987654321	Abner	M	1942-02-28	Spouse
Joyce	A	English	453453453	1972-07-31	5631 Rice,Houston,TX	F	25000.00	333445555	5	333445555	Theodore	M	1983-10-25	Son
Joyce	A	English	453453453	1972-07-31	5631 Rice,Houston,TX	F	25000.00	333445555	5	333445555	Joy	F	1958-05-03	Spouse
Joyce	A	English	453453453	1972-07-31	5631 Rice,Houston,TX	F	25000.00	333445555	5	333445555	Alice	F	1986-04-05	Daught
Joyce	A	English	453453453	1972-07-31	5631 Rice,Houston,TX	F	25000.00	333445555	5	123456789	Michael	M	1988-01-04	Son
Joyce	A	English	453453453	1972-07-31	5631 Rice,Houston,TX	F	25000.00	333445555	5	123456789	Elizabeth	F	1967-05-05	Spouse
Joyce	A	English	453453453	1972-07-31	5631 Rice,Houston,TX	F	25000.00	333445555	5	123456789	Alice	F	1988-12-30	Daught
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000.00	333445555	5	987654321	Abner	M	1942-02-28	Spouse
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000.00	333445555	5	333445555	Theodore	M	1983-10-25	Son

The resultant relation will have 56 rows, 15 columns.

Generally, CROSS PRODUCT is not a meaningful operation, can become meaningful when followed by other operations

Example (not meaningful):

$\text{EMP_DEPENDENTS} \leftarrow \text{EMPLOYEE} \times \text{DEPENDENT}$

EMP_DEPENDENTS will contain every combination of EMPNAMES and DEPENDENT whether or not they are actually related

- To keep only combinations where the DEPENDENT is related to the EMPLOYEE, we add a SELECT operation as follows:

Example (meaningful):

$EMP_DEPENDENTS \leftarrow EMPLOYEE \times DEPENDENT$

$ACTUAL_DEPS \leftarrow \sigma_{SSN=ESSN}(EMP_DEPENDENTS)$

RESULT (ACTUAL_DEPS) will now contains employees and their dependents information

Fname	Minit	Lname	Ssn	Bdate	Address	Gender	Salary	Super_ssn	Dno	Essn	Dependent_name	Gender	Bdate	Relation
John	B	Smith	123456789	1965-01-09	731 Fondren,Houston,TX	M	30000.00	888665555	5	123456789	Alice	F	1988-12-30	Daughter
John	B	Smith	123456789	1965-01-09	731 Fondren,Houston,TX	M	30000.00	888665555	5	123456789	Elizabeth	F	1967-05-05	Spouse
John	B	Smith	123456789	1965-01-09	731 Fondren,Houston,TX	M	30000.00	888665555	5	123456789	Michael	M	1988-01-04	Son
Franklin	T	Wong	333445555	1955-12-08	638 voss,Houston,TX	M	40000.00	888665555	5	333445555	Alice	F	1986-04-05	Daughter
Franklin	T	Wong	333445555	1955-12-08	638 voss,Houston,TX	M	40000.00	888665555	5	333445555	Joy	F	1958-05-03	Spouse
Franklin	T	Wong	333445555	1955-12-08	638 voss,Houston,TX	M	40000.00	888665555	5	333445555	Theodore	M	1983-10-25	Son
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire,Tx	F	43000.00	333445555	4	987654321	Abner	M	1942-02-28	Spouse

The issue with Cartesian Product is that it returns **all possible combinations of rows** from both tables.

This results in **redundant and meaningless combinations** unless you filter them using a condition.(as shown in previous slide: $\sigma_{SSN=ESSN}$)

A **natural or join**, on the other hand, **performs a Cartesian Product internally and automatically filters rows by matching columns with the same name and compatible data types**, thus ensuring that there are no redundancies in the results.

- The **join** operation allows us to combine a select operation and a Cartesian-Product operation into a single operation.
- Consider relations $r (R)$ and $s (S)$
- Let “theta” be a predicate on attributes in the schema $R \cup S$. The join operation $r \bowtie_{\theta} s$ is defined as follows:

$$r \bowtie_{\theta} s = \sigma_{\theta} (r \times s)$$

- Thus

$$\sigma_{instructor.id = teaches.id} (instructor \times teaches)$$

- Can equivalently be written as

$$instructor \bowtie_{instructor.id = teaches.id} teaches.$$

Example: Write an relational algebraic query to retrieve the details of Managers(Fname, Name) and their managing department name.

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

$DEPT_MGR \leftarrow DEPARTMENT \bowtie_{Mgr_ssn=Ssn} EMPLOYEE$
 $RESULT \leftarrow \pi_{Dname, Lname, Fname}(DEPT_MGR)$

DEPT_MGR

Dname	Dnumber	Mgr_ssn	...	Fname	Minit	Lname	Ssn	...
Research	5	333445555	...	Franklin	T	Wong	333445555	...
Administration	4	987654321	...	Jennifer	S	Wallace	987654321	...
Headquarters	1	888665555	...	James	E	Borg	888665555	...

Q:

Consider the employee database of Figure 2.17. Give an expression in the relational algebra to express each of the following queries:

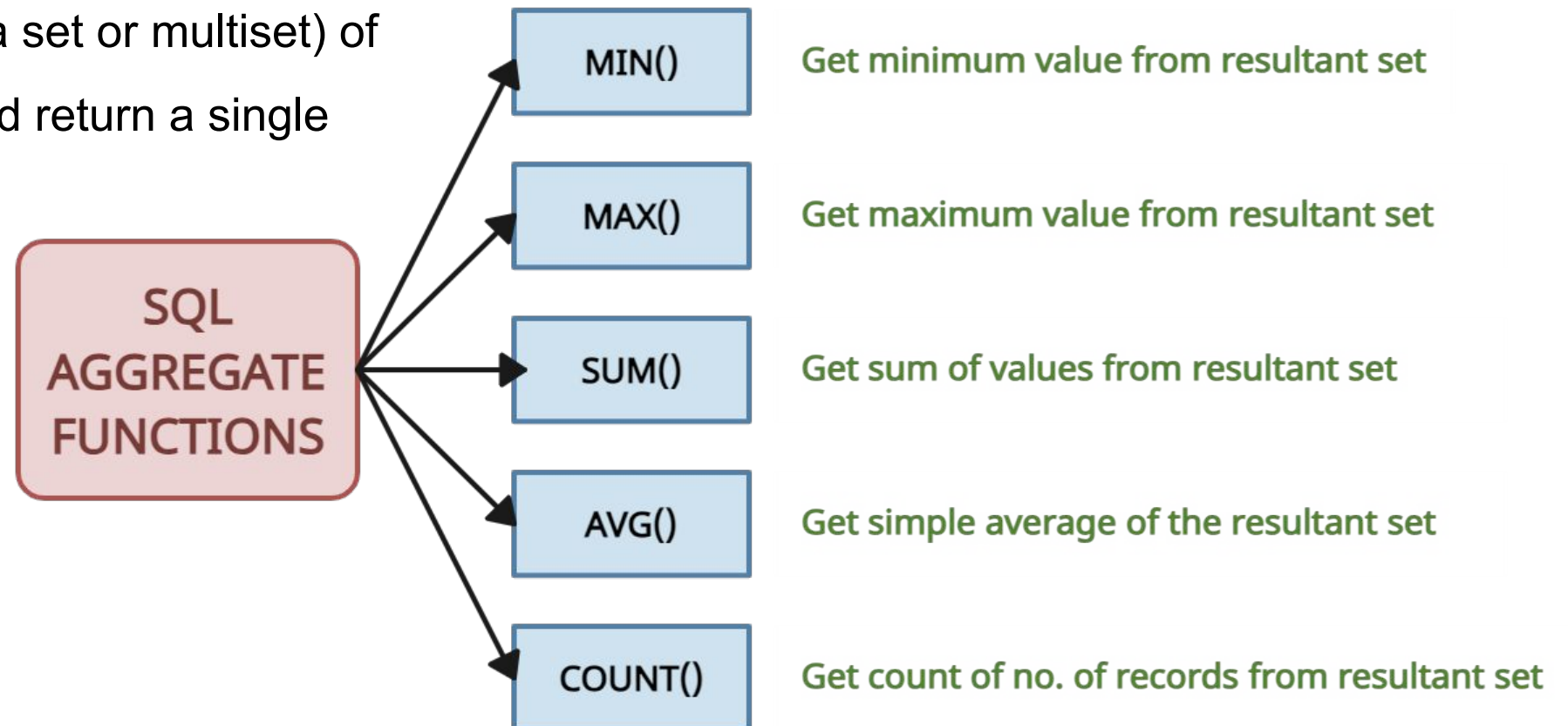
- Find the name of each employee who lives in city “Miami”.
- Find the name of each employee whose salary is greater than \$100000.
- Find the name of each employee who lives in “Miami” and whose salary is greater than \$100000.

employee (*person_name*, *street*, *city*)
works (*person_name*, *company_name*, *salary*)
company (*company_name*, *city*)

Figure 2.17 Employee database.

Aggregate Functions and types:

Aggregate functions are functions that take a collection (a set or multiset) of values as input and return a single value.



Database Management Systems

Aggregate Functions and Grouping



- A type of request that cannot be expressed in basic relational algebra is to specify mathematical **aggregate functions** on collections of values from the database.
- Examples of such functions include retrieving the average or total salary of all employees or the total number of employee tuples.
- These functions are used in simple statistical queries that summarize information from the database tuples.
- Common functions applied to collections of numeric values include SUM, AVERAGE, MAXIMUM, and MINIMUM.
- The COUNT function is used for counting tuples or values.

Note: count just counts the number of rows, without removing duplicates

Database Management Systems

Aggregate Functions and Grouping



Use of the Aggregate Functional operation $\mathcal{F}$

- $\mathcal{F}_{\text{MAX Salary}}$ (EMPLOYEE) retrieves the maximum salary value from the EMPLOYEE relation.
- $\mathcal{F}_{\text{MIN Salary}}$ (EMPLOYEE) retrieves the minimum salary value from the EMPLOYEE relation.
- $\mathcal{F}_{\text{SUM Salary}}$ (EMPLOYEE) retrieves the sum of salary values from the EMPLOYEE relation.
- $\mathcal{F}_{\text{COUNT SSN, AVERAGE Salary}}$ (EMPLOYEE) computes the count (number) of employees & their average salary.

Average-

- The function is used to calculate the average value of the numeric type.
- average function returns the average of all non-Null values.

Ex:

A	B	C
α	α	7
α	β	7
β	β	3
β	β	10

Output: $\mathcal{F}_{average(C)}(R)=27/4=6.75$

Database Management Systems

Count Function

Count:

Count function is used to count the number of tuples in a relation.

It can work on both numeric and non-numeric data types.

Ex:

A	B	C
α	α	7
α	β	7
β	β	3
β	β	10

Output: $\mathcal{F}_{count(*)}(E)=4$

4 is the number of tuples in the above table

Note: $\mathcal{F}$ or g can be used

Sum-

- Sum function is used to calculate the sum of all selected columns.
- It works on numeric fields only.

Ex:

■ Relation r :

A	B	C
α	α	7
α	β	7
β	β	3
β	β	10

$g_{\text{sum}(c)}(r)$

sum-C
27

Minimum-

- MIN function is used to find the minimum value of a certain column.
- This function determines the smallest value of all selected values of a column.

Ex-

$$g_{\min(C)}(R)=3$$

Maximum-

- MAX function is used to find the maximum value of a certain column.
- This function determines the largest value of all selected values of a column.

Ex-

$$g_{\max(C)}(R)=10$$

A	B	C
α	α	7
α	β	7
β	β	3
β	β	10

Database Management Systems

Aggregate Functions and Grouping

Consider the table:

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Database Management Systems

Aggregate Functions and Grouping



Example:

- 1) Find the sum of the salaries of all employees, the maximum salary, the minimum salary, and the average salary.

Query:

$\rho_{R(\text{Total_Sal, Highest_Sal, Lowest_sal, Average_Sal})} \approx (\text{SUM}(\text{Salary}), \text{MAX}(\text{Salary}), \text{MIN}(\text{Salary}), \text{AVG}(\text{Salary})) (\text{EMPLOYEE})$

Output:

Total_Sal	Highest_Sal	Lowest_Sal	Average_Sal
345930.00	55000.00	25000.00	38436.666667

Database Management Systems

Aggregate Functions and Grouping



Grouping can be combined with Aggregate Functions

Example: For each department, retrieve the DNO, COUNT SSN, and AVERAGE SALARY

- A variation of aggregate operation $\mathcal{F}$ allows this:
 - Grouping attribute placed to left of symbol
 - Aggregate functions to right of symbol
 - $\text{DNO } \mathcal{F} \text{ COUNT SSN, AVERAGE Salary (EMPLOYEE)}$
- Above operation groups employees by DNO (department number) and computes the count of employees and average salary per department

- In many cases aggregate functions can be applied to subgroups of data.
Example: Calculate average salary for each department.
- In these cases data is divided into non-overlapping groups based on specified attributes.
- Each group (partition) will consist of the tuples that have the same value of some attribute(s), called the grouping attribute(s).
- We can then apply the function to each such group independently to produce summary information about each group. SQL has a GROUP BY clause for this purpose.
- The GROUP BY clause specifies the grouping attributes, which should also appear in the SELECT clause, so that the value resulting from applying each aggregate function to a group of tuples appears along with the value of the grouping attribute(s).

Database Management Systems

Aggregate Functions and Grouping

Example: find the average salary of employees in each department

Query:

$\rho_{R(Dno, count(*), Avg(Salary))} (Dno \bowtie (Dno, COUNT(*), AVG(Salary)) (EMPLOYEE))$

Output:

Fname	Minit	Lname	<u>Ssn</u>	...	Salary	Super_ssn	Dno
John	B	Smith	123456789		30000	333445555	5
Franklin	T	Wong	333445555		40000	888665555	5
Ramesh	K	Narayan	666884444		38000	333445555	5
Joyce	A	English	453453453	...	25000	333445555	5
Alicia	J	Zelaya	999887777		25000	987654321	4
Jennifer	S	Wallace	987654321		43000	888665555	4
Ahmad	V	Jabbar	987987987		25000	987654321	4
James	E	Bong	888665555		55000	NULL	1

Grouping EMPLOYEE tuples by the value of Dno

Dno	Count (*)	Avg (Salary)
5	4	33250
4	3	31000
1	1	55000

Result of Q24

Database Management Systems

Aggregate Functions and Grouping



Exercise:

Write Relational Algebra Statements to:

- a) Retrieve each department number, the number of employees in the department, and their average salary, while renaming the resulting attributes.
- b) Show the department number, the total number of employees, and the average salary of employees in that department, for each department.
- c) Find the total number of employees and the average salary across the entire company.

Database Management Systems

Aggregate Functions and Grouping

Answers:

a. $\rho_{R(\text{Dno, No_of_employees, Average_sal})}(\text{Dno} \bowtie \text{COUNT Ssn, AVERAGE Salary (EMPLOYEE)})$.

b. $\text{Dno} \bowtie \text{Count ssn, Average_salary (EMPLOYEE)}$.

c. $\bowtie \text{COUNT ssn, AVERAGE Salary (EMPLOYEE)}$.

Output:

R

(a)

Dno	No_of_employees	Average_sal
5	4	33250
4	3	31000
1	1	55000

(b)

Dno	Count_ssn	Average_salary
5	4	33250
4	3	31000
1	1	55000

(c)

Count_ssn	Average_salary
8	35125

Database Management Systems

Questions and Answers



Fill in the Blanks:

1) The union operation allows us to combine two relations (r and s). The notation used for union is _____.

Ans: $r \cup s$

2) For the union operation $r \cup s$ to be valid, r and s must have the same _____.

Ans: arity (same number of attributes)

3) In the union operation, duplicate tuples are _____.

Ans: eliminated

4) The set-intersection operation allows us to find tuples that are in both input relations. The notation used for intersection is _____.

Ans: $r \cap s$

Database Management Systems

Questions and Answers



Fill in the Blanks:

5) The set-difference operation allows us to find tuples that are in one relation but not in another. The notation used for set-difference is _____.

Ans: $r-s$

6) The number of attributes in a relation is referred to as its _____.

Ans: arity

7) The two queries that give the same result on any database but are written differently in relational algebra are called _____ queries.

Ans: equivalent

8) In the UNION operation, the two operand relations must be "type _____".

Ans: compatible

Can you write the intersection operation using union and set difference operations ?

INTERSECTION can be expressed in terms of union and set difference as follows:

$$R \cap S = ((R \cup S) - (R - S)) - (S - R)$$



THANK YOU

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